

Predicting weather-related power outages in large scale distribution grids with deep learning ensembles

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ABSTRACT

Weather events are primarily contributors to electrical supply disruptions, prompting the need to accurately forecast these weather-related power outages. This paper focuses on predicting daily reported incidences in electrical grids within specific regions, leveraging weather conditions as specific predictive variables. An optimization-based feature selection approach has been considered for selecting the optimal Reanalysis node locations used as predictors. To overcome the data imbalance challenge and enhance prediction accuracy, we propose a Deep Learning-based (DL) ensemble algorithm. Five distinct DL architectures are considered, generating multiple individual learners with randomly selected hyperparameters. Diversity is ensured by training each model with slightly different randomly sampled data. Three information fusion techniques construct the final ensemble models. The proposed approach has been successfully evaluated in predicting real daily reported incidences in distribution lines across two Spanish provinces, Valencia and Albacete, achieving an accurate prediction of days with extreme incidences while maintaining a good overall performance and a low rate of false alarms. The top-performing models using this methodology achieve detection rates of 38% and 73%, with false alarm rates of 1% and 3%, respectively. This approach not only enhances prediction accuracy compared to individual learners but also improves the generalization ability and robustness of standalone DL models. Additionally, it effectively reduces the inherent overfitting of these methods, removing the necessity for a complex hyperparameter selection process.

1. Introduction

A robust and reliable electricity delivery is paramount nowadays. Uninterrupted access to electricity is a keystone for any nation's economic advancement. A disruption in the supply has essential direct and indirect financial consequences, frequently surpassing the intrinsic value of the electricity not supplied, particularly in the context of significant blackout events [1]. Furthermore, the increasing interdependencies among various infrastructure components, such as communication networks, financial markets, and transportation systems, amplify the potential for electric blackouts to impact other vital infrastructures [2,3].

Understanding the historical patterns of failures in electricity grids is crucial for policymakers when prioritizing improvements across various infrastructures [4,5]. While power failures can result from diverse causes such as equipment malfunctions, human errors, or earthquakes, events associated with extreme weather represent the leading

cause of electrical supply disruptions [1,6,7]. These high-impact, low-probability (HILPs) events [8,9] are noted for their rapid and devastating nature, causing damage to multiple components across extensive areas. Consequently, these events incur substantial costs for repairs and lead to significant economic losses [10,11].

In this context, providing accurate forecasting of power outages derived from disturbances in the electrical grid plays an important role, enabling the preventive reinforcement of the affected areas, thereby minimizing the ensuing damage and inconvenience caused by these disruptions in the power supply. Nevertheless, due to the inherent uncertainties associated with most of the factors responsible for the majority of power distribution line disturbances, such as extreme weather events, vegetation, animals, etc., model-based approaches struggle to deliver precise outage predictions in many cases [12]. In contrast, data driven Machine Learning (ML) methods can better explore the intricate

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relationships among predictive factors contributing to power outages and the target output.

Diverse applications emerge in the literature in outage forecasting and vulnerability identification. Some of them are focused on predicting the duration of a specific power outage [13,14]. At the same time, other studies deal with predicting the risk associated with a particular power outage [15,16]. In the realm of assessing the impact of extreme meteorological events on power distribution lines, numerous studies in the literature aim to forecast the damage incurred or the number of outages caused by these severe weather events [17–27].

This paper focuses on predicting the number of incidences reported in the electrical grid within a specific region, utilizing current weather conditions as predictive variables. Similar approaches in the literature can address classification or regression problems. In the context of classification, in [28], authors consider the problem of predicting power outages based on the current weather conditions. They deploy novel graph neural networks to process weather measurements to adequately determine the power outage likelihood. In [29] a 3-dimensional SVM model is implemented to predict the outage of power system components caused by extreme events. On the other hand, similar works address this issue as a regression problem: in [30], the authors review the existing power outage models, highlighting their strengths and limitations. These existing outage models were validated using power outages from multiple regions assolated by different hurricanes, including hurricanes Harvey (2017), Michael (2018), and Isaias (2020), in over 1900 US cities. They focused on two response variables: the number of outages, equivalent to the number of customers without power in a city, and the fraction of customers without power. In [31], an ensemble learning approach based on a boosting algorithm, AdaBoost, is proposed for estimating power outages induced by weather conditions. The findings indicate that the proposed ensemble model outperforms both regression and neural network models, offering more accurate results. In [32] a methodology to forecast weather-related power distribution outages one day-ahead hourly is presented. Also, a solution to address the data imbalance issue is proposed, where only a tiny portion of the data represents the hours impacted by outages in the form of a weighted logistic regression model. The weights for outage and non-outage hours are determined by the reciprocals of their corresponding number of hours. Two case studies using data from a small electric utility company in the United States were analyzed to demonstrate the proposed model's effectiveness.

This paper addresses the prediction challenge of forecasting the number of daily reported incidences in a specific region, framed as a regression problem. Our approach begins by utilizing an evolutionary algorithm to extract optimal reanalysis node locations, determining the meteorological variables to be employed as predictors. The main forecasting difficulty lies in managing highly unbalanced databases, where instances corresponding to days with many reported incidences constitute a minimal percentage of the overall data. Despite their limited occurrence, these instances are crucial for achieving accurate predictions. We propose a DL-based ensemble algorithm to overcome the challenge posed by data imbalance, which hinders the correct prediction of high-incidence days by base models. Specifically, five distinct DL architectures are considered, generating multiple individual learners. Hyperparameters, encompassing data preprocessing, model architecture, and training procedures, are randomly selected within predefined discrete ranges for each model. Also, a random subset of the dataset is employed for the training of each individual learner to ensure diversity within the ensemble. Subsequently, different techniques are applied to construct the ensemble models. The impact of the filtering process and the percentage of individual models entering the ensemble, named as elitism level, is also investigated. This DL ensemble approach has been successfully applied to a different problem of low-visibility events prediction problem related to fog incidents [33]. Similar architectures composed of DL hybrid ensembles of prediction

models have been successfully applied recently to the field of power load forecasting [34,35].

The key contributions of this work include: (i) the implementation of an evolutionary algorithm to optimally select reanalysis node locations and determine the most relevant meteorological predictors; (ii) the development of an ensemble learning approach leveraging diversity among individual DL learners to mitigate overfitting and enhance generalization in the extreme events predictions; and (iii) an in-depth analysis of ensemble construction strategies, evaluating the impact of filtering mechanisms and elitism levels on predictive performance. The effectiveness of the proposed methodology in predicting extreme weather-related power outages is evaluated through two real-world case studies, specifically addressing the daily number of reported incidences in distribution lines across two Spanish provinces: Valencia and Albacete. The comprehensive assessment includes the ensemble models' performance analysis, considering various factors such as elitism level or ensemble composition.

The rest of the paper is organized as follows: First, Section 2 describes the data used and the problem particularities. Then, Section 3 describes in detail the methodologies used for carrying out the daily incidences forecast. Details on the experimental work and the achieved results are presented in Section 4. Finally, some discussion, conclusions and final outlook are given in Section 5.

2. Data and problem particularities

The primary objective of this study is the prediction of the number of incidences impacting electrical distribution networks due to extreme meteorological events. Our approach formulates this challenge as a regression problem, where the target variable is the daily count of incidences occurring in a specific region.

To initiate our investigation, we have access to two databases spanning the years 2016 to 2021. These databases detail the number of incidences affecting distribution lines in two Spanish provinces, Valencia and Albacete. Each incident is associated with a specific date and a categorized cause. Fig. 1 illustrates the geographic locations of these regions, providing a visual context for our dataset. Albacete and Valencia are regions in eastern Spain, with different characteristics. The province of Valencia is a coastal region. It includes the city of Valencia, the third largest city in Spain with a population of around 800,000 inhabitants. The area's electricity demand is high, due to its urbanization and industrial activities. In contrast, Albacete is a province located inland, within the Castilla-La Mancha region. It has a population of around 385,000 inhabitants, and its economy is mainly based on agriculture, with a smaller urban footprint. The city of Albacete is the largest urban area in the province, but the region has smaller towns and villages than Valencia, so the energy consumption is also smaller.

Figs. 2 and 3 show the time series of daily incidences reported in Albacete and Valencia, respectively. For confidentiality purposes, data is normalized between 0 and 1. Thus, 1 corresponds to the maximum number of daily incidents in each database. From these figures, two main inferences can be drawn: (1) the data present an undeniable seasonality with a yearly periodicity, which is further illustrated in Fig. 4, revealing that the summer and autumn months are those with the highest number of daily incidences, and (2) some abnormally high incidence reporting days appear in the time series; these days are precisely the ones with the most significant interest in the problem since they represent days in which the distribution lines have suffered outages or more severe problems.

In order to properly assess the performance of the proposed methodologies in the forecasting of those days with a significantly high number of incidences, two different objectives have been established in the resolution of the problem:

- Regression problem: the aim will be to predict, with the lowest possible error, the number of daily incidents reported in the province.

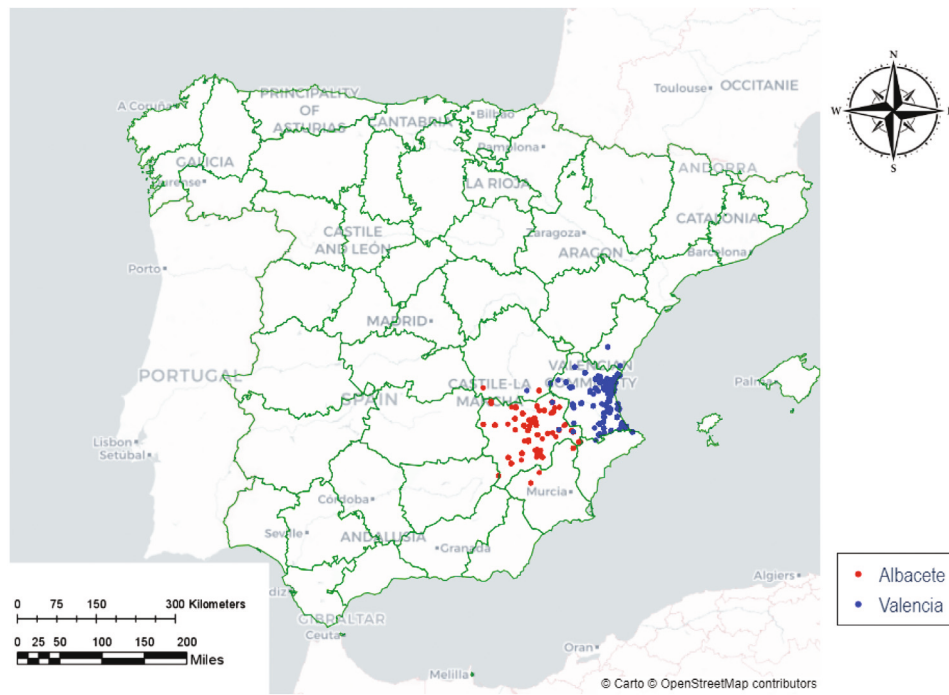


Fig. 1. Geographic distribution of reported incidences in Albacete and Valencia regions (Spain), from 2016 to 2021.

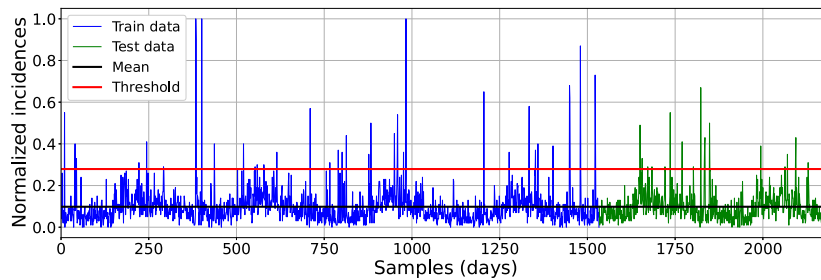


Fig. 2. Daily incidences reported in the province of Albacete.

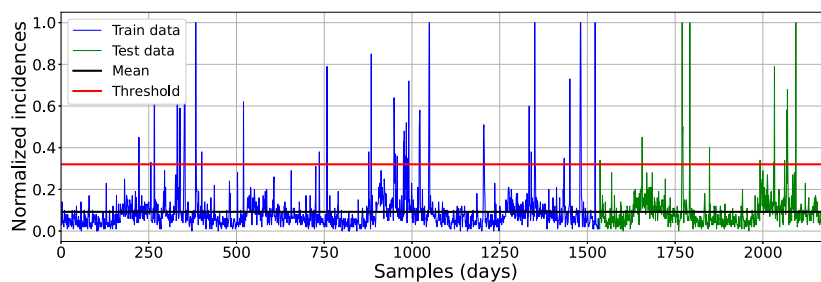


Fig. 3. Daily incidences reported in the province of Valencia.

- Classification problem: a threshold corresponding to a high-incidences-day is established, and classification metrics are used accordingly to assess whether the models can classify a given day as ‘extreme’ or ‘non-extreme’. As shown in Figs. 2 and 3, this threshold has been fixed at percentile 30 of the data following experts advice.

The challenge of this forecasting problem lies in dealing with highly unbalanced databases since the number of instances with a high number of reported incidences represents a minimum percentage of the total data, being precisely these data those that are most crucial to estimate properly. Specifically, the percentage of days with several incidences

above the threshold corresponds to 2.28% for the province of Albacete and 3.06% for the province of Valencia.

2.1. Predictive variables

The forecasting methodology outlined in this paper relies on meteorological data obtained from the reanalysis project ERA5 [36], employing nodes strategically positioned across the geographical area under investigation. ERA5 offers hourly data on various meteorological parameters, including temperature, pressure, precipitation, and snow-fall, with a spatial resolution of 0.25 degrees longitude and latitude between grids.

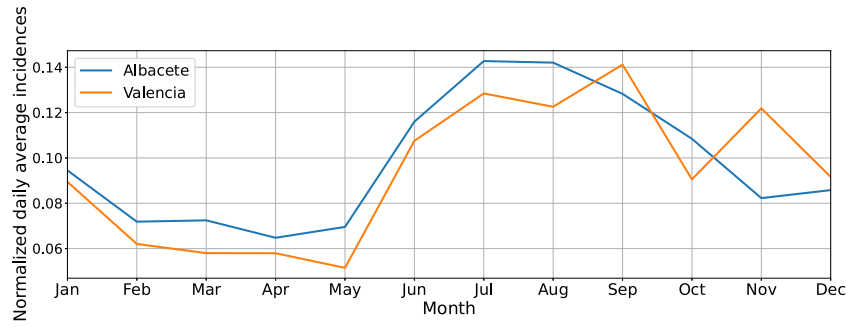


Fig. 4. Monthly distribution of daily reported incidences in the provinces of Albacete and Valencia.

Table 1

Predictive variables considered at each node from the ERA-5 reanalysis.

Variable name	ERA5 variable	Daily aggregation
d2m	2 m dewpoint temperature	Average value
t2m	2 m temperature	Average value
ws	10 m wind speed module	Average value
tp	Total precipitation	Maximum value
tcw	Total column rain water	Maximum value
csf	Convective snowfall	Maximum value

To address the challenge of forecasting the daily count of reported incidences, the chosen predictive variables encompass temperature, humidity, wind speed across different components at varying heights and diverse precipitation and snowfall metrics. Table 1 enumerates the six selected predictive variables per node. These variables have been resampled to a daily scale, with the aggregation method for hourly values specified in the table for each variable.

To manage computational costs associated with an increased number of predictor variables (6 for each node), we have limited the number of reanalysis nodes selected for the forecasting problem to four. An evolutionary algorithm has been employed to identify the nodes that contribute the most relevant information to the prediction systems. This algorithm, detailed in Section 3.1, is designed to extract the optimal combination of nodes for the cases of Albacete and Valencia. The application of this algorithm facilitates the selection of a subset of nodes that efficiently captures essential information for accurate predictions, mitigating the computational burden associated with an exhaustive use of all available nodes.

2.2. Error metrics

For assessing the performance of the prediction models, different evaluation metrics have been employed, from generic regression error metrics to specific metrics that reflect the model's performance in the prediction of extreme events.

2.2.1. Mean Absolute Error (MAE)

First, a common indicator for regression problems, Mean Absolute Error (MAE), has been considered:

$$MAE = \frac{1}{N} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (1)$$

where $\hat{y}$ represents predicted values (provided by the model) and y are the actual values. The subscript i refers to a single sample $y_i = y[i]$.

2.2.2. Extreme Events Mean Absolute Error (EEMAE)

The focus of this work is the precise prediction of extreme events, and as such, the evaluation is specifically directed towards forecasting these critical points. To initiate this assessment, it is necessary to establish a threshold beyond which a point is considered an extreme event. Figs. 2 and 3 present the target time series, segmented into

training (70%) and testing (30%) datasets. These figures illustrate the threshold to differentiate extreme data (outliers) from non-extreme data (non-outliers), set at 30 reported incidences after consultation with experts. To measure the performance of the forecasting models on these outlier data, an indicator focused on the prediction of extreme events from a regression perspective, named Extreme Events Mean Absolute Error (EEMAE) [37–39], has been used, which corresponds to the MAE (Eq. (1)) calculation on these values.

First, an indicator function that selects only the data points where the absolute value of the true event exceeds the threshold (Th in the Equation) is defined:

$$I(y_i, T) = \begin{cases} 1 & \text{if } |y_i| \geq T \\ 0 & \text{if } |y_i| < T \end{cases} \quad (2)$$

Then, the EEMAE formula can be written as:

$$EEMAE = \frac{1}{N} \sum_{i=1}^N I(y_i, Th) |y_i - \hat{y}_i| \quad (3)$$

2.2.3. True Positive Rate (TPR), False Positive Rate (FPR), G-mean (G-mean) and F1-score

A potential limitation of the previously presented metric, EEMAE, is its exclusive focus on the prediction of extreme values without penalizing situations where the model incorrectly predicts an extreme event when the actual value is normal (False Positive). Such false alarms could result in severe economic damages, necessitating their minimization. To address this concern, four well-established binary classification error metrics have been employed: True Positive Rate (TPR), False Positive Rate (FPR), Geometric Mean (G-mean) and F1-score. These metrics provide a comprehensive evaluation by accounting for correct and incorrect predictions, offering insights into the model's ability to balance sensitivity and specificity, particularly concerning extreme event predictions.

3. Proposed methodology

This section describes in detail the different methodologies used in the forecasting work. First, the Differential Evolution optimization algorithm used for the reanalysis node selection is described in Section 3.1. Then, the proposed DL ensemble approach is further described in Section 3.2. Finally, the ML and DL methods used to assess the performance of the proposed methodology are described in Section 3.3.

3.1. Differential evolution

Differential Evolution (DE) [40] consists of a stochastic, population-based and direct searching method designed for global optimization problems. DE uses a real encoding method, and its survival strategy is based on the simple differential mutation operation, and one-to-one competition reduces the complexity of the genetic operation. Therefore, solving the optimization problem in a complex environment is suitable, which cannot be solved using conventional mathematical programming

methods. It keeps a population with N_p individuals, where every individual within the population stands for a possible solution to the problem. Individuals are represented by a vector $X_{i,g}$, where $i = 1, \dots, N_p$ and g refer to the index of the generation. A DE cycle consists of three consecutive steps: mutation, crossover and selection.

3.2. Deep learning ensembles

Ensemble learning consists in the integration of knowledge for several individual learners from heterogeneous or homogeneous models to obtain better generalization performance [41]. It is based on the use of voting schemes in an adaptive way to overcome model overfitting and initialization sensitivity [42].

The proposed approach for this unbalanced prediction problem relies on exploring different ensemble and information fusion techniques applied over individual deep models for time series prediction. Therefore, the strengths of methods including bagging [43], stacking [44] and super learners [45] are combined to enhance the prediction of individual regression models. Specifically, for each of the 5 deep methods under study, including five multivariate time-series DL methods (Recurrent Neural Networks (RNNs) [46], Gated Recurrent Units (GRUs) [47], Long Short-Term Memory (LSTM) [48], 1-D Convolutional Neural Networks (1D-CNN) [49] and 2-D Convolutional Neural Networks (2D-CNN) [50]), several individual models are created by randomly sampling the training data from the complete dataset. Hyperparameters for each learner are selected randomly within a defined discrete range. As a result, each individual model exhibits distinct characteristics, which collectively enhance the overall ensemble. This ensemble framework has been previously used for successfully solving a problem of visibility prediction [33].

3.2.1. Data splitting

The original dataset is split into 50% for training, 20% for validation, and 30% for testing. The first portion is used to train the individual deep regression models, while the validation set is employed for training and configuring the ensemble learning models. Finally, the test data is reserved for evaluating the performance of all developed models. Given the nature of time-series data, the temporal order of the samples is preserved in the splitting procedure.

3.2.2. Hyperparameters selection

The parameters considered in the creation on individual models, together with its potential values, are listed in Table 2. This tunable hyperparameters are divided into three groups: those related to the data preprocessing, those regarding the DL models structure and those concerning its training procedure. The specific values of this parameter are selected to ensure a diverse range of complexity among the individual models included in the ensemble.

3.2.3. Data preprocessing

This stage involves transforming the time-series predictive training data into three-dimensional tensors structured as: $[samples \times timestep \times variables]$. This time-series tensors contains daily-sampled weather information, with their length determined by the hyperparameter *sequence length* defined in Table 2. Additionally, this preprocessing step involved assigning each sample, represented as a matrix of dimensions $[timestep \times variables]$, the corresponding target variable value $y(t)$, for a sample with time values between $(t - timestep)$ and t , since no time prediction horizon has been set in the problem.

Table 2
Hyperparameters considered.

	Parameter	Values
Data prep	Sequence length	2,3,4,5
	Number of layers	1,2,3
Model structure	Number of neurons	16,32,64,128,256,516
	Number of dense layers	1,2,3
	Number of dense neurons	16,32
	Kernel size	2,3,4,5,6
	Activation function	ReLU, SeLU, ELu
Training parameters	Loss function	MAE, MSE, Huber Loss, Log cosh
	Optimizer	Adam, RMS Prop, Adadelta, Adagrad, Adamax, Ftrl, Nadam
	Batch size	16,32,64,128
	Patience (early stopping)	25,50,75,100,125,150
	Maximum epochs	200,400,600, ..., 5800,6000

3.2.4. Creation and training of the models

For each of the five DL architectures considered, the following procedure is performed:

1. Training subsets are created by sampling a random 80% of the training data samples that has been preprocessed.
2. Hyperparameters are selected by a random choice procedure, and individual learners are subsequently created.
3. Training of individual learners is performed, randomly selecting the training parameters. 10% validation split from the training data is performed without shuffling for monitoring the validation loss and stopping the training process when no improvements are obtained for x consecutive epochs, where x is the selected patience parameter.

Finally, N individual models are obtained for each DL methodology (with $N = 100$ in the experimental work performed). These models are referred to as individual learners. The total of 500 learners ($100learners \times 5methods$), are saved and subsequently used to create the ensemble models.

3.2.5. Ensemble learning

After the training of individual learners is performed, their predictions are combined using various ensemble learning techniques to produce the final predicted value. This process is illustrated in Fig. 5. Initially, the performance of each individual model is evaluated using the validation data. Then, the top-performing models are selected using several selection criteria: varying both the percentage of models selected (the best 4%, 10%, 25%, 50% or the 100% of the models) and the selection metric (MAE, EEMAE). Three information fusion methodologies are utilized to develop the ensemble models, which are described in detail in this section. For each information fusion approach, a total of 10 ensemble models are evaluated (2 metrics $\times$ 5 elitism levels).

Unweighted model averaging. The most commonly used approach in the ensemble learning field consists of computing the unweighted average of the individual predictions [41]. Given that DL architectures typically exhibit high variance and low bias, this simple averaging technique enhances generalization performance by reducing the variance among the models.

Weighted model averaging. Unweighted model averaging, although being a suitable option when the individual regressors exhibit similar performance [51], it can lead to suboptimal results in cases where the ensemble includes heterogeneous base models, as weaker learners may negatively impact overall performance. In a weighted model averaging approach, higher consideration is given to the best-performing learners.

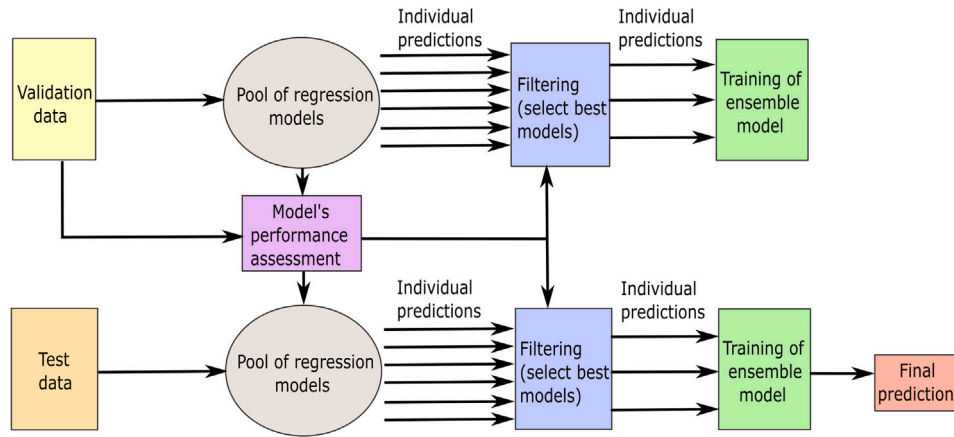


Fig. 5. Flowchart of the implemented ensemble procedure.

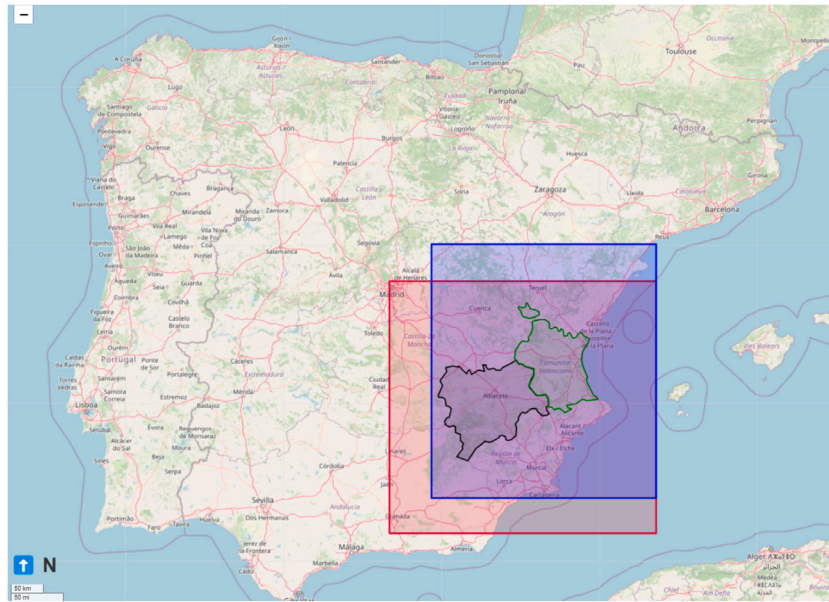


Fig. 6. Geographic limits established for the DE algorithm. Red square represents the limits for the case of Albacete and blue one the case of Valencia. Contour of both regions are plotted. Black contour represent the limits of Albacete province and green contour of Valencia.

To achieve this, each individual model performance is assessed using the validation data. An error value ($\mathcal{E}_i$) is assigned to each learner. Next, the weights of each model (ω_i) are calculated by subtracting the normalized validation error from 1 (as shown in Eq. (4)). This approach ensures that the best-performing model receives a weight of 1 ($\omega_i = 1$), while the worst-performing model has a weight of 0 ($\omega_i = 0$). Finally, the predictions from base learners are weighted according to their corresponding weights, and the ensemble output is calculated by averaging these weighted values (Eq. (5)).

$$\omega_i = 1 - \frac{(\mathcal{E}_i - \min(\mathcal{E}))}{(\max(\mathcal{E}) - \min(\mathcal{E}))} \quad (4)$$

$$\hat{y}_{ensemble} = \frac{\sum_{i=1}^N \hat{y}_i \cdot \omega_i}{\sum_{i=1}^N \omega_i} \quad (5)$$

Super learning. Super learning [52] is an ensemble learning technique that is based on the weight optimization of individual learners. A loss function is minimize for this purpose. In this work, a sample split strategy is implemented: First, the individual learners are fitted on the training data, then, the weights of each learner are optimized using the validation data. A Deep Fully Connected Neural Network with a single hidden is employed as super learner model, similarly to [53]. The

number of neurons has been set equal to twice the number of inputs (this being the number of individual learners filtered). The Tangent Hyperbolic (tanh) function was used as activation function, Adam was chosen as the optimization algorithm, and the training parameters were configured as follows: the batch size was set to 32, patience was established at 75 epochs without improvement, validation split without shuffling was set as 10% and the validation loss metric used was either MAE or MSE depending on the filtering criteria selected for recruiting the individual learners that formed part of the ensemble.

3.3. Algorithms for comparison

Six classical ML methods for regression problems, together with five state-of-the-art DL techniques, have been employed for compare and validate the results obtained with the proposed ensemble methodology. The ML methods include: Linear Regression (LR) [54], Regression Trees (RT) [55], Random Forest (RF) [56], Support Vector Regression (SVR) [57], Multilayer Perceptron (MLP) [58] and Extreme Learning Machine (ELM) [59]. On the other hand, the DL techniques include: Residual Network (ResNet) [60], InceptionTime [61], Time Series Transformer (TST) [62], Multilevel wavelet decomposition network (mWDN) [63] and Autoregressive integrated moving average

Table 3

Geographic limits established for the DE algorithm.

Limit	Albacete	Valencia
North	40.50 N	41.00 N
South	37.00 N	37.50 N
West	3.75 W	3.00 W
East	1.00 W	1.00 E

Table 4

Geographic nodes selected for Albacete province.

Node number	Latitude	Longitude
1	39.50 N	3.45 W
2	38.25 N	2.70 W
3	39.00 N	3.45 W
4	38.25 N	2.45 W

Table 5

Geographic nodes selected for Valencia province.

Node number	Latitude	Longitude
1	39.50 N	3.00 W
2	38.75 N	0.50 W
3	39.00 N	2.00 W
4	38.75 N	0.75 W

with explanatory variable (ARIMAX) [64]. The hyperparameters for these methods were selected through a random search approach. In this process, the training data was split, reserving the last 20% for validation and identifying the optimal hyperparameter combination.

4. Experimental work

This section shows the experimental work performed in the forecasting of the daily number of reported incidences in the provinces of Albacete (Section 4.2) and Valencia (Section 4.3). In first place, the optimization-based node selection for extracting the predictive variables in each case is described in Section 4.1.

4.1. Node selection

The DE optimization algorithm was performed in order to extract the combination of four nodes that provides the best forecasting results on the validation data. In first place, Table 3 and Fig. 6 show the geographic limits established for each case, these limits are set seeking for a slightly larger area than the province under consideration.

Then, the fitness function for the optimization algorithm was defined as the sum of MAE error of the different ML algorithms (Eq. (6)). This has been done looking for the combination of nodes that provides better overall results.

$$F_i = MAE_{LR} + MAE_{RT} + MAE_{RF} + MAE_{SVR} + MAE_{ELM} + MAE_{MLP} \quad (6)$$

Finally, nodes returned by the optimization algorithm are shown in Tables 4 and 5 for the cases of Albacete and Valencia, respectively, and depicted in Fig. 7, where it can be observed that the most representative nodes for each case do not necessarily have to be in the corresponding province.

4.2. Daily reported incidences forecasting: Albacete

Results for the applied methodology are presented in detail in this section, using as target variable the daily number of reported incidences in the province of Albacete.

4.2.1. Individual DL models

Initially, 100 individual learners were trained for each DL technique, with distinct sets of hyperparameters randomly assigned and utilizing slightly varied portions of the training data. The outcomes of these individual learners, measured in terms of MAE and EEMAE, are visually represented in Fig. 8.

4.2.2. Ensemble DL models

Subsequently, these individual models were amalgamated into an ensemble using three distinct criteria (SL, UA, and WA) and varying elitism levels (top 4%, 10%, 25%, 50%, and 100%) based on two metrics: MAE and EEMAE. The outcomes of these ensemble models are illustrated in Fig. 9, demonstrating superior performance across both metrics compared to the individual models, and revealing a discernible Pareto front. Notably, among the specific ensemble strategies, the most favorable results (characterized by lower MAE and EEMAE) were consistently attained with the UA and WA strategies. In these figures, we also present the outcomes achieved with the ML and DL algorithms designated as benchmarks, observing that the ensemble models proposed, specially the ones build based on the UA and WA criteria obtained the most satisfactory results.

Finally, in Table 6, the additional error metrics for the optimal ensemble models are presented. Selecting the ensemble models that perform best in the metric they were specifically designed. Conclusively, ensembles fashioned according to the EEMAE metric exhibit highly satisfactory results, showcasing a commendable equilibrium between extreme and non-extreme predictions while maintaining a low incidence of false alarms. Concerning the benchmark algorithms, it is noteworthy that only the RF model attains highly competitive results, aligning with its presence in the Pareto front as depicted in Fig. 9. Nevertheless, this model exhibits less favorable outcomes in terms of TPR and G-Mean. Consequently, it can be inferred that the most successful model in this scenario is the RNN ensemble constructed based on the EEMAE metric. The time series predictions of this ensemble are illustrated in Fig. 10.

4.3. Daily reported incidences forecasting: Valencia

Results for the applied ensemble methodology are presented in this section for the province of Valencia (East of Spain).

Fig. 11 illustrates the comparison between individual DL learners, focusing on MAE and EEMAE errors, juxtaposed against the performance of ensemble models and benchmark algorithms. Notably, the ensemble learners designed with UA and WA strategies showcase a well-balanced and satisfactory outcome across the two evaluated metrics, presenting a Pareto front similar to the previous scenario.

The performance of the models regarding the other error metrics can be consulted in Table 7. Notably, the individual DL models demonstrate superior results in terms of extreme event prediction (EEMAE and TPR). However, this advantage is counterbalanced by a higher occurrence of false alarms (FPR) and a less accurate average prediction (MAE). This inherent challenge is effectively addressed through the ensemble approach, resulting in well-balanced outcomes and models proficient in forecasting both extreme and non-extreme events. Finally, Fig. 12 presents the time series prediction of the most satisfactory model, the LSTM ensemble based on the EEMAE metric.

5. Conclusions

This paper proposes a DL ensemble-based approach tailored for forecasting daily reported incidences in electrical distribution networks within Spain, focusing on the provinces of Albacete and Valencia. However, the methodologies and conclusions derived from this study are universally applicable to diverse territories.

The ability to accurately forecast the number of incidences in electrical distribution networks carries several significant implications,



Fig. 7. Geographic nodes selected by the DE algorithm for the cases of Albacete (red) and Valencia (blue).

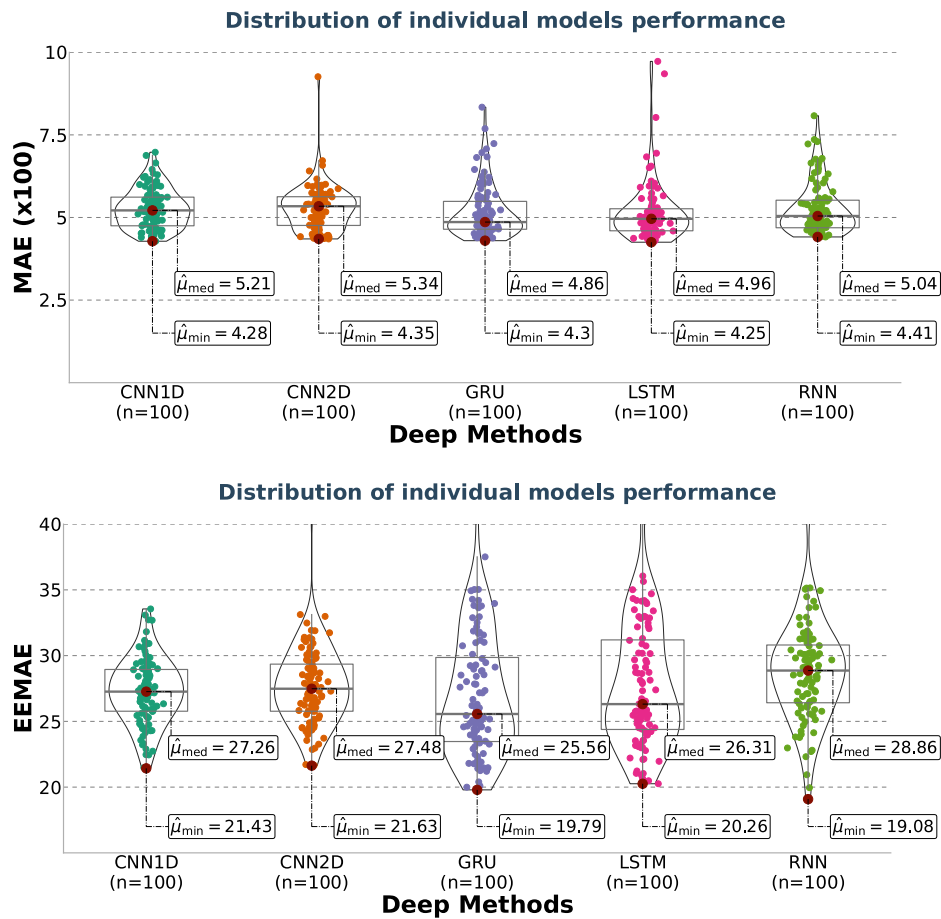


Fig. 8. Distribution plot of the individual learners performance (in terms of MAE and EEMAE) for Albacete database. The violin plot shows the values that are more repeated for each of the DL methodologies assessed. Median and minimum values are also indicated for each case.

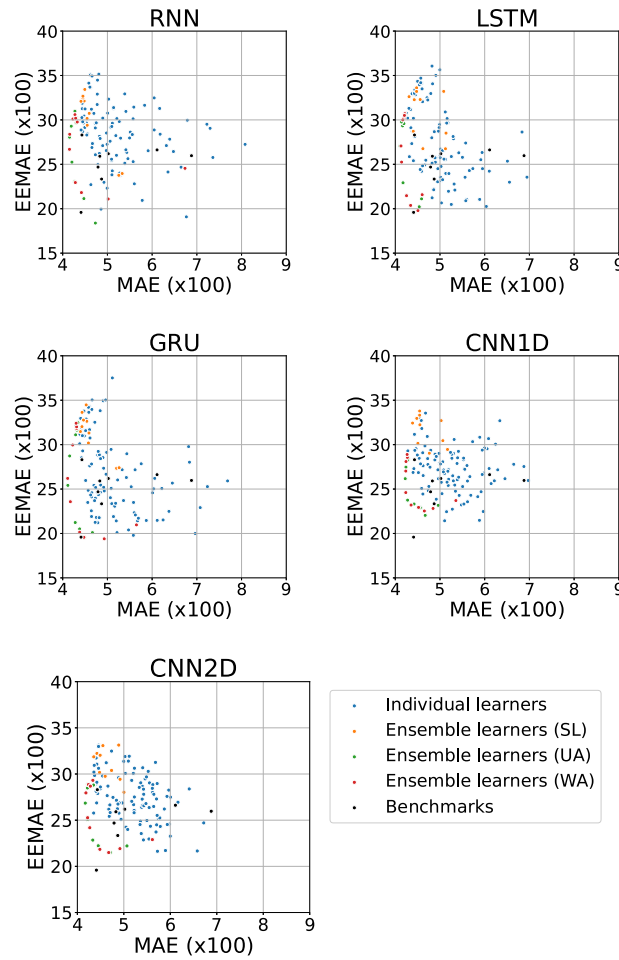


Fig. 9. Comparison of the performance of individual learners versus ensemble learners and benchmark algorithms in terms of MAE and EEMAE, for the RNN, LSTM, GRU, CNN1D and CNN2D methodologies, in Albacete database.

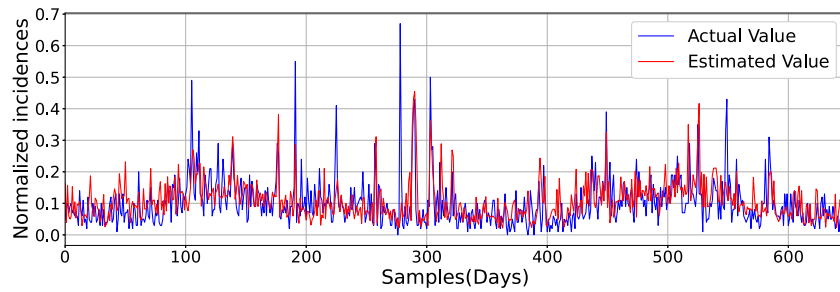


Fig. 10. RNN forecasting time series of extreme values (red) vs. actual extreme values (blue).

with broad-reaching impacts on various stakeholders and aspects of the energy sector, including operational efficiency, customer satisfaction, emergency response, and long-term planning. It also contributes to the overall reliability, resilience, and sustainability of energy infrastructure. However, this poses an intrinsic problem of data imbalance, which implies finding a balance between achieving accurate predictions for days with a high number of incidents and ensuring accurate predictions for the remaining days, while minimizing false alarms.

To overcome this challenge, a DL-based ensemble approach has been adopted, where five distinct DL architectures are considered, generating multiple individual learners with randomly selected hyperparameters, and trained with slightly different randomly sampled data. Subsequently, three information fusion techniques are applied to construct the ensemble models. The results obtained by this method offered a balanced performance in extreme detection rates, while keeping

low the false alarm rate. The best performing models considering this methodology report results with 38% and 73% of extreme detection, with 1% and 3% of false alarms rate, for the cases of Albacete and Valencia, respectively.

Despite the promising results, the proposed method has some limitations. First, the model's performance is influenced by the availability and quality of historical incident data, which may limit its applicability in regions with sparse or incomplete datasets. Second, the DL models used lack explainability, making it difficult to interpret the factors contributing to predictions and, consequently, challenging their acceptance by operators and decision-makers. Finally, incorporating additional exogenous predictive variables, such as geospatial data, or network maintenance indicators, could enhance the model's predictive capabilities and robustness across different operational scenarios.

Table 6
Error metrics for albacete database.

		MAE(x100)	EEMAE(x100)	TPR	FPR	G-Mean	F1-score
ML methods	LR	4.79	24.69	0.23	0.00	0.48	0.32
	RT	6.11	26.62	0.38	0.03	0.61	0.29
	RF	4.41	19.59	0.31	0.01	0.55	0.40
	SVR	4.87	23.34	0.08	0.02	0.27	0.08
	ELM	6.88	25.97	0.54	0.04	0.72	0.26
	MLP	4.43	28.30	0.15	0.01	0.39	0.27
DL methods	ResNet	4.83	25.91	0.08	0.01	0.28	0.27
	InceptionTime	5.02	26.19	0.38	0.02	0.61	0.25
	TST	5.63	24.33	0.23	0.02	0.48	0.22
	mWDN	4.65	23.77	0.23	0.02	0.48	0.00
	Arimax	4.66	30.66	0.00	0.00	0.00	0.00
Best ind learners (MAE)	RNN	4.41	28.86	0.00	0.00	0.00	0.00
	LSTM	4.25	30.74	0.00	0.00	0.00	0.00
	GRU	4.30	32.19	0.00	0.00	0.00	0.00
	CNN1D	4.28	29.30	0.00	0.00	0.00	0.00
	CNN2D	4.35	25.77	0.15	0.00	0.39	0.25
Best ind learners (EEMAE)	RNN	6.77	19.08	0.38	0.03	0.61	0.28
	LSTM	6.04	20.26	0.38	0.03	0.61	0.26
	GRU	5.58	19.79	0.23	0.02	0.47	0.19
	CNN1D	5.10	21.43	0.23	0.01	0.48	0.30
	CNN2D	5.72	21.63	0.23	0.03	0.47	0.18
Ensemble metric MAE	RNN	4.15	26.69	0.00	0.00	0.00	0.00
	LSTM	4.11	27.05	0.00	0.00	0.00	0.00
	GRU	4.11	26.20	0.00	0.00	0.00	0.00
	CNN1D	4.22	26.17	0.08	0.00	0.28	0.13
	CNN2D	4.17	26.85	0.08	0.00	0.28	0.14
Ensemble metric EEMAE	RNN	4.73	18.39	0.38	0.01	0.62	0.40
	LSTM	4.51	19.79	0.31	0.01	0.55	0.35
	GRU	4.93	19.40	0.31	0.02	0.55	0.27
	CNN1D	4.67	22.04	0.15	0.01	0.39	0.20
	CNN2D	4.67	21.50	0.23	0.01	0.48	0.27

Table 7
Error metrics for valencia database.

		MAE(x100)	EEMAE(x100)	TPR	FPR	G-Mean	F1-score
ML methods	LR	5.41	28.10	0.44	0.02	0.65	0.39
	RT	7.04	38.88	0.19	0.04	0.42	0.19
	RF	4.91	28.11	0.44	0.03	0.65	0.36
	SVR	5.28	29.81	0.56	0.03	0.74	0.42
	ELM	7.74	23.97	0.56	0.06	0.73	0.30
	MLP	5.22	29.14	0.50	0.02	0.70	0.40
DL Methods	ResNet	6.91	36.53	0.33	0.03	0.57	0.36
	InceptionTime	6.64	31.33	0.53	0.03	0.72	0.35
	TST	8.01	35.74	0.20	0.03	0.44	0.11
	mWDN	5.52	28.83	0.47	0.03	0.67	0.39
	Arimax	4.87	39.47	0.12	0.01	0.35	0.18
Best ind learners (MAE)	RNN	4.55	42.96	0.00	0.00	0.00	0.00
	LSTM	4.38	44.59	0.00	0.00	0.00	0.00
	GRU	4.40	42.29	0.00	0.00	0.00	0.00
	CNN1D	4.54	41.00	0.00	0.00	0.00	0.00
	CNN2D	4.64	39.18	0.13	0.00	0.36	0.20
Best ind learners (EEMAE)	RNN	7.63	16.13	0.73	0.06	0.83	0.34
	LSTM	6.05	21.38	0.67	0.03	0.80	0.45
	GRU	5.58	18.70	0.80	0.05	0.87	0.41
	CNN1D	6.05	21.08	0.67	0.03	0.80	0.44
	CNN2D	6.60	19.84	0.73	0.06	0.83	0.35
Ensemble metric: MAE	RNN	4.43	41.81	0.00	0.00	0.00	0.00
	LSTM	4.40	36.22	0.20	0.00	0.45	0.33
	GRU	4.44	42.95	0.00	0.00	0.00	0.00
	CNN1D	4.47	42.38	0.00	0.00	0.00	0.00
	CNN2D	4.47	43.04	0.00	0.00	0.00	0.00
Ensemble metric: EEMAE	RNN	5.49	23.56	0.67	0.04	0.80	0.41
	LSTM	4.92	23.51	0.73	0.03	0.84	0.51
	GRU	5.18	24.13	0.60	0.03	0.76	0.41
	CNN1D	5.92	24.00	0.67	0.03	0.80	0.43
	CNN2D	5.37	23.97	0.60	0.03	0.76	0.39

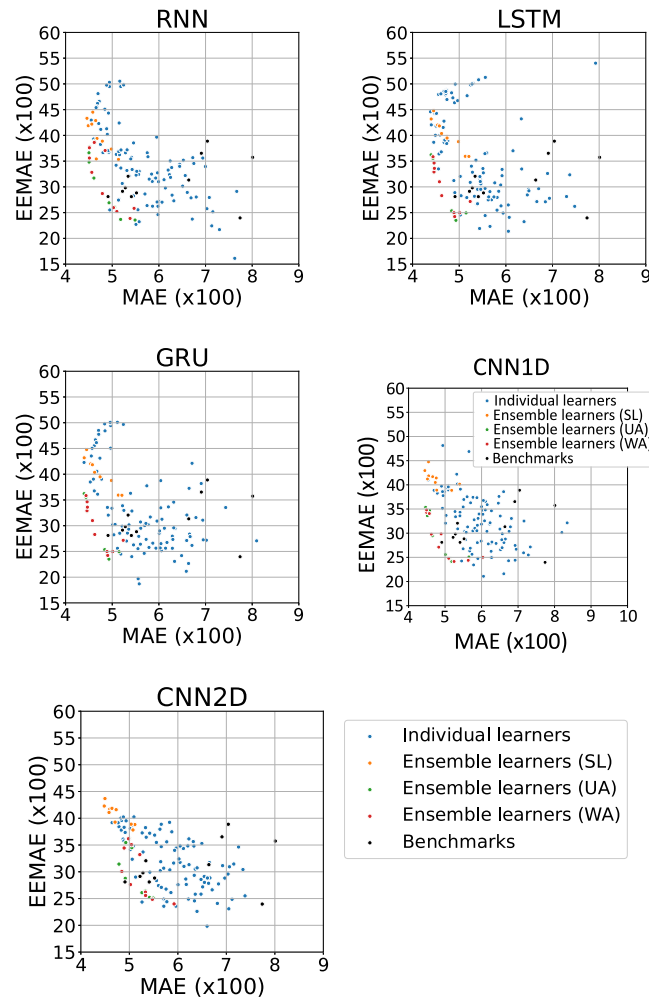


Fig. 11. Comparison of the performance of individual learners versus ensemble learners and benchmark algorithms in terms of MAE and EEMAE, for the RNN, LSTM, GRU, CNN1D and CNN2D methodologies, in Valencia database.

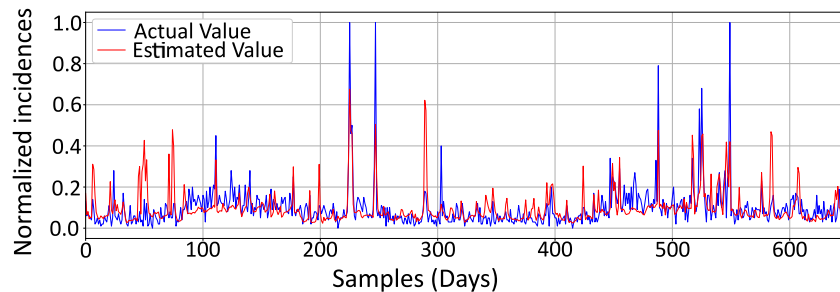


Fig. 12. LSTM based-ensemble (build in accordance to EEMAE validation error) forecasting time series (red) vs. actual values (blue).

To address these limitations, future research will explore strategies to improve model interpretability, such as using explainable artificial intelligence (XAI) techniques. Additionally, the integration of new exogenous data sources will be analyzed to enrich the model and enhance its predictive capacity. Adaptive learning approaches will also be considered to mitigate dependence on historical data and improve the model's applicability in contexts with limited information.

In summary, this study not only reveals the intrinsic challenge of data imbalance in predicting incidences within electrical distribution networks, but also underscores the critical need for sophisticated methodologies to achieve an optimal balance. Simultaneously, our proposed DL ensemble methodology, when applied to two locations tested,

demonstrates notable success. This methodology not only improves predictions compared to individual learners, but also enhances the generalization capability and robustness of single DL models. Furthermore, it effectively mitigates the inherent overfitting associated with these methods, eliminating the need for a complex hyperparameter selection procedure.

Finally, note that this model has been recently included in operation at Iberdrola (multinational company owning wind farms in Spain). The model provides alerts which help correctly manage incidents in the energy transportation infrastructures. Iberdrola feeds the model with the weather forecasting for the next day, and it provides alerts of possible high number of incident in the network, helping managing them and reducing risk of power outages.

CRedit authorship contribution statement

L. Prieto-Godino: Writing – original draft, Supervision, Methodology, Investigation. **C. Peláez-Rodríguez:** Writing – original draft, Visualization, Software, Methodology, Investigation. **J. Pérez-Aracil:** Writing – review & editing, Validation, Software, Resources, Conceptualization. **J. Pastor-Soriano:** Writing – review & editing, Investigation, Formal analysis, Conceptualization. **S. Salcedo-Sanz:** Writing – review & editing, Investigation, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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